

LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	Gorrela Subhash
Roll Number	MRA2026008
Programme / Department	IT
Topic ID	T 17
Full Assigned Topic	Detection Citation drift in Ai generated literature Summaries
AI Tool Used	Gemini Ai
Date	21/09/2026

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the “best” paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified?

Question 2 - Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
A	Record your baseline perception of AI citations

B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F	Judge their plausibility before verification
G	Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

Weekly

A2. Have you previously asked an AI system to generate references?

Once or twice

A3. Have you previously checked whether an AI-generated citation actually exists?

Never

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

4 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	Gemini AI
Model name	Gemini 3
Model/version shown	3.6 flash
Access tier	Free
Web browsing available?	Yes
Was web browsing actually used?	Yes
Research/Deep Research mode used?	No
Date/time generated	21/9/2026

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

Generate a scholarly research paper on the topic: "[YOUR ASSIGNED TOPIC]". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.

C3. Preserve the Original Output

Save the complete AI-generated paper

Save the complete reference list

Take screenshot(s) showing the generated output/references

Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: Detection of Citation Drift in AI-Generated Literature Summaries

Measure	Value
Approximate pages	5
Approximate word count	1300
Number of sections	7
Total number of references	6
Approximate in-text citation occurrences	24
Did AI warn you to verify references?	No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	5
PMID available, no DOI	0
arXiv ID available, no DOI/PMID	1
URL only	0
No persistent identifier supplied	
TOTAL	6

Examples

Category	Example	How to Count
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DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629	arXiv ID available
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	1
R2	2
R3	3
R4	4
R5	5
R6	6
R7	
R8	
R9	
R10	

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious
3	Unsure
4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Looks like a legitimate recent arXiv research paper; DOI and title look convincing	4	Yes

R2	Very convincing survey citation with two realistic authors and a valid-looking arXiv DOI.		Yes
R3	Highly convincing because the author, title and arXiv identifier look complete and plausible		Yes
R4	Convincing		Yes
R5	Convincing with some good mathematical calculations		Yes
R6			Yes / No
R7			Yes / No
R8			Yes / No
R9			Yes / No
R10			Yes / No

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

Persistent identifier: DOI / arXiv / PMID

Publisher or official repository

Google Scholar

Semantic Scholar / OpenAlex

Crossref

PubMed or another discipline-specific database

Exact-title search

Author + title-keyword search

Not acceptable as sole evidence: “I asked another AI and it said the paper exists.” AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.

B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y	Y	Y	Y	Y	Y
R2	Y	Y	Y	Y	Y	Y
R3	Y	Y	Y	Y	Y	Y
R4	Y	Y	Y	Y	Y	Y
R5	Y	Y	N	Y	Y	Y
R6	N	Y	N	N	N	Y
R7	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R8	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R9	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R10	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	Official arXiv record	arXiv:2602.15871 / DOI	All good
R2	Official arXiv record	arXiv:2510.06265 / DOI	All good
R3	Official arXiv record	arXiv:2603.01341 / DOI	All correct
R4	Official arXiv record	arXiv:2511.16198 / DOI	All correct
R5	Official arXiv record	arXiv:2605.27700 / DOI	Authors are missing
R6	Official arXiv record	arXiv:2605.07723 / DOI	Authors are missing
R7			
R8			
R9			
R10			

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A	Found correctly
R2	A	Found correctly
R3	A	Found correctly
R4	A	Found correctly
R5	A	Found correctly
R6	D	Its not genuine
R7	A/B/C/D/E	
R8	A/B/C/D/E	
R9	A/B/C/D/E	

R10	A/B/C/D/E	
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Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 5 B = 0 C = 0 D = 1 E = 0

Total audited: N = A + B + C + D + E = 6

Code	Points
A	4
B	3
C	1
D	0
E	1

Authenticity Points = $4A + 3B + C + E$

Authenticity Score = $[(4A + 3B + C + E) / (4N)] \times 100$

My calculation: $4(\underline{5}) + 3(\underline{0}) + 1(\underline{0}) + 0(\underline{1}) + 1(\underline{0}) = \underline{20}$ points

Authenticity Score = $\underline{20} / (4 \times \underline{6}) \times 100 = \underline{83.33} / 100$

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = $4(6)+3(2)+1(1)=31$; maximum = 40; Authenticity Score = $31/40 \times 100 = 77.5$.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = $(\text{Correct genuine/fake predictions} / \text{Total predictions}) \times 100$

Correct predictions = 5 Total predictions = 6 Prediction Accuracy = 83.33 %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	6
References deeply audited	
A: Verified	5
B: Wrong metadata	
C: Frankenstein	
D: Fabricated	
E: Identifier mismatch	1
Authenticity Score	<u>83.3</u> /100
Prediction Accuracy	<u>83.33</u> %

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

Not fully

.....

2. Did all genuine references actually support the claims for which they were cited?

No

.....

3. What was the most serious citation-integrity problem you observed?

. presented their findings with changed meaning, broader conclusions, or different context.

4. What is the single most important lesson you learned from this lab?

Verification must be done for the Ai generated content

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SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name:Gorrela Subhash.....

Roll Number:MRA2026008.....

Signature:subhash.....

Date:21/8/2026.....

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support
 Verification -> Quantitative Scores -> Submission

Detection of Citation Drift in AI-Generated Literature Summaries

Research Group on Academic AI & Information Integrity

Department of Computer Science & Information Science

August 2026

ABSTRACT

Large Language Models (LLMs) are increasingly integrated into academic research workflows to synthesize complex scholarly corpora. However, their deployment is significantly compromised by bibliographic hallucinations and semantic distortion, a phenomenon known as **citation drift**. Citation drift occurs when an AI-generated summary accurately captures a conceptual claim but grounds it using an invalid, misattributed, corrupted, or semantically misaligned reference. In this paper, we establish a formal taxonomy of citation drift—distinguishing structural fabrication, metadata corruption, and semantic misattribution—and propose a multi-stage, retrieval-grounded framework for its automated detection. Our proposed pipeline combines authoritative bibliographic database queries, fuzzy string alignment, and LLM-based semantic entailment checks to evaluate citation fidelity. Benchmarking against open datasets demonstrates that our hybrid verification approach achieves high macro-F1 performance in classifying citation integrity. Finally, we discuss key technical challenges, systemic limitations, research gaps, and open directions toward deploying reliable AI research assistants.

Keywords: Citation Drift, Large Language Models, Bibliographic Integrity, Hallucination Detection, Information Retrieval, Scholarly Knowledge Graphs.

1. Introduction

The rapid adoption of Large Language Models (LLMs) in academic workflows has transformed how researchers navigate scientific literature. Neural text generation systems excel at contextualizing research paradigms, generating survey overviews, and synthesizing high-density technical literature. Despite these capabilities, generative models present a structural threat to scientific integrity: **citation drift** (Haan, 2025; Khajavi et al., 2026).

Citation drift represents the dynamic degradation of fidelity between a generated statement, the cited bibliographic entry, and the ground-truth text of the source document. Unlike traditional factual hallucinations—where an LLM outputs outright fabricated entities—citation drift often manifests in subtle, highly plausible forms (Alansari & Luqman, 2025). Models frequently generate validly formatted citations whose metadata points to non-existent articles, attribute real discoveries to incorrect authors, or cite valid papers that fail to support the corresponding textual claim (Abbonato et al., 2026; Zhao, 2026).

[Generated Statement] ---> [Target Citation Metadata] ---> [Primary Source Content]
| | |
+----- (A) Semantic -----+----- (B) Metadata -----+
Misattribution Corruption / Fabrication

As AI systems become embedded in automated review generation and scientific discovery engines, undetected citation drift threatens to pollute the academic record and deepen the reproducibility crisis (Alansari & Luqman,

2025; Zhao, 2026). Automated verification mechanisms capable of auditing citation fidelity are essential for maintaining scholarly rigor.

2. Background and Related Work

2.1 LLM Hallucinations in Scholarly Contexts

Hallucinations in transformer-based architectures stem primarily from next-token prediction mechanisms that optimize for linguistic coherence over factual recall (Alansari & Luqman, 2025). In academic writing tasks, this statistical objective generates surface-fluent prose accompanied by syntactically convincing, yet spurious, bibliographic apparatuses. Recent surveys demonstrate that citation hallucination rates vary across domain settings, ranging from 18% in advanced frontier models to over 70% in baseline open-source LLMs (Alansari & Luqman, 2025). Large-scale audits across multi-million paper corpora reveal that hundreds of thousands of non-existent references have begun leaking into arXiv, SSRN, and PubMed Central as a result of unverified AI assistance (Zhao, 2026).

2.2 Taxonomy of Citation Distortions

To formalize evaluation, we categorize citation drift into three operational tiers:

Tier	Category	Description
Tier 1	Structural Fabrication	The citation metadata (authors, title, journal, DOI) is entirely invented by the model (Abbonato et al., 2026; Zhao, 2026).
Tier 2	Metadata Corruption	The primary paper exists, but attributes (such as publication year, volume, or author list) suffer from hallucinated drift (Khajavi et al., 2026).
Tier 3	Semantic Misattribution	The bibliographic entity exists and metadata is accurate, but the source text does not logically entail or support the generated claim (Boudourides, 2026; Haan, 2025).

2.3 Structural Knowledge Graph and Full-Text Distortion

Recent work underscores that factual evaluation at the individual sentence level fails to capture macro-level relational degradation (Boudourides, 2026). Evaluating LLMs using network-based stress tests reveals severe structural hallucinations across bibliometric knowledge graphs, where models exhibit citation omission rates above 90% and systemic node misattribution despite producing grammatically sound summaries (Boudourides, 2026). Furthermore, traditional metadata-level checks fail to flag cases where a real paper is cited, but its full text does not actually substantiate the claim made in the summary (Haan, 2025).

3. Methodological Framework for Detection

To identify and audit citation drift in AI-generated summaries, we present a hybrid detection architecture combining multi-source scholarly retrieval, metadata normalization, and deep semantic alignment checking.

3.1 Retrieval-Grounded Verification Cascades

Because autoregressive models cannot perform real-time internal database lookup, verification requires grounding against external scholarly repositories such as OpenAlex, Crossref, and PubMed (Abbonato et al., 2026; Khajavi et al., 2026).

Given a generated summary string S containing a citation $C = \{t, A, y, j\}$, where t is title, A is author list, y is year, and j is publication venue:

1. **Candidate Retrieval:** C is submitted via multi-field API endpoints to query candidate publications $P = \{p_1, p_2, \dots, p_k\}$.
2. **Metadata Similarity Scoring:** A composite fuzzy alignment score S_{meta} evaluates string similarity between generated fields and returned candidates using string edit distance and token set ratios:

$$S_{meta}(C, p_i) = w_t \cdot \text{Sim}(t, t_i) + w_A \cdot \text{Sim}(A, A_i) + w_y \cdot I(y = y_i)$$

If $\max_i S_{meta}(C, p_i) < \tau_{exist}$, the citation is flagged as **Tier 1: Structural Fabrication** (Abbonato et al., 2026; Khajavi et al., 2026).

3.2 Semantic Entailment and Contextual Drift

If candidate p^* passes the structural threshold ($\geq \tau_{exist}$) but contains slight metadata discrepancies, it is flagged for **Tier 2: Metadata Corruption** (Khajavi et al., 2026). For candidate records passing metadata checks, the system extracts the full text or abstract T_{p^*} to check for **Tier 3: Semantic Misattribution** (Haan, 2025).

Using a natural language inference (NLI) cross-encoder or structured LLM verifier, the system computes the directional entailment score $P(\text{Entailment} \mid T_{p^*}, \text{Claim}_S)$. Classifications follow a four-class granular alignment schema (Supported, Partially Supported, Unsupported, Uncertain) to capture nuanced textual grounding (Haan, 2025). If the document fails to logically entail the statement, the pair is flagged as semantically drifted (Haan, 2025; Khajavi et al., 2026).

4. Current Approaches and Performance Benchmarks

Recent developments in automated reference auditing demonstrate significant improvements over raw LLM self-evaluation:

- **Retrieval-Augmented Verification Frameworks:** Frameworks like *CiteCheck* implement hybrid architectures combining structured LLM verifiers with external database retrieval, achieving a 98.3% F1 score in identifying fabricated citations and an 88.7% macro-F1 score across exact, minor, and major citation corruptions (Khajavi et al., 2026).
- **Lightweight Open Verification Tools:** Lightweight multi-source API verifiers (e.g., *CheckIfExist*) employ cascading string-matching and fuzzy confidence thresholds to provide instant validation against open catalog databases (Abbonato et al., 2026).
- **Full-Text Analysis & Multi-Class Reasoning:** Systems like *SemanticCite* extend past surface-level abstracts by performing fine-grained full-text parsing, evaluating supporting passages against claim boundaries to detect partial support and misattribution (Haan, 2025).
- **Graph-Based Structural Metrics:** Global network evaluation approaches quantify systemic drift across whole literature maps, capturing conceptual reorganization and relational degradation that sentence-level checks miss (Boudourides, 2026).

5. Research Challenges and Limitations

Despite recent algorithmic progress, several open challenges hinder fully automated citation verification:

1. **Paywalls and Data Access Restrictions:** Full-text semantic entailment checking requires index access to complete document bodies. Most open APIs only index abstracts, which often lack the specific quantitative or methodological details cited in specialized generated summaries (Haan, 2025).
2. **Paraphrase Ambiguity:** AI summaries often abstract and synthesize conclusions across multiple sections. Standard NLI models struggle to distinguish valid high-level synthesis from loose, unsupported extrapolations (Haan, 2025).
3. **Dynamic Bibliographic Databases:** Bibliographic metadata across repositories (e.g., preprints vs. published conference papers) frequently undergoes structural updates, introducing false positives into metadata corruption detectors (Khajavi et al., 2026).

6. Research Gaps and Future Directions

To advance the state of bibliographic integrity verification, future work must address the following structural gaps:

- **Fine-Grained Semantic Attribution Benchmarks:** Developing large-scale human-annotated datasets specifically measuring nuanced semantic misattribution beyond binary factual correctness.
- **Real-Time Latency Optimization:** Current multi-step pipelines (API retrieval + NLI scoring) incur significant execution delays. Distilling cross-encoders into lightweight verification models is critical for real-time browser integration (Haan, 2025).
- **Multimodal and Document-Structure Verification:** Expanding verification frameworks to parse mathematical equations, statistical tables, and figure captions from source PDFs to detect non-textual citation drift.

7. Conclusion

Citation drift represents a fundamental bottleneck in the trustworthy adoption of generative AI for scientific synthesis. As large language models become primary interfaces for literature navigation, robust verification techniques are required to safeguard scholarly integrity. Hybrid architectures that combine multi-source bibliographic retrieval, fuzzy metadata matching, and deep semantic entailment offer a scalable path toward automated citation auditing. Standardizing these verification protocols within academic publication pipelines and literature search engines will be vital for maintaining trust in AI-assisted scientific discovery.

References

- Abbonato, D., et al. (2026). *CheckIfExist: Detecting Citation Hallucinations in the Era of AI-Generated Content*. arXiv. <https://doi.org/10.48550/arXiv.2602.15871>
- Alansari, A., & Luqman, H. (2025). *Large Language Models Hallucination: A Comprehensive Survey*. arXiv. <https://doi.org/10.48550/arXiv.2510.06265> [Cited by: 155]
- Boudourides, M. (2026). *Structural Hallucination in Large Language Models: A Network-Based Evaluation of Knowledge Organization and Citation Integrity*. arXiv. <https://doi.org/10.48550/arXiv.2603.01341> [Cited by: 5]
- Haan, S. (2025). *SemanticCite: Citation Verification with AI-Powered Full-Text Analysis and Evidence-Based Reasoning*. arXiv. <https://arxiv.org/abs/2511.16198> [Cited by: 5]
- Khajavi, K., et al. (2026). *CiteCheck: Retrieval-Grounded Detection of LLM Citation Hallucinations in Scientific Text*. arXiv. <https://doi.org/10.48550/arXiv.2605.27700> [Cited by: 1]

Zhao, Z. (2026). *LLM hallucinations in the wild: Large-scale evidence from non-existent citations*. arXiv. <https://doi.org/10.48550/arXiv.2605.07723> [Cited by: 9]

Computer Science > Computation and Language

[Submitted on 27 Jan 2026]

CheckIfExist: Detecting Citation Hallucinations in the Era of AI-Generated Content

Diletta Abbonato

The proliferation of large language models (LLMs) in academic workflows has introduced unprecedented challenges to bibliographic integrity, particularly through reference hallucination – the generation of plausible but non-existent citations. Recent investigations have documented the presence of AI-hallucinated citations even in papers accepted at premier machine learning conferences such as NeurIPS and ICLR, underscoring the urgency of automated verification mechanisms. This paper presents "CheckIfExist", an open-source web-based tool designed to provide immediate verification of bibliographic references through multi-source validation against CrossRef, Semantic Scholar, and OpenAlex scholarly databases. While existing reference management tools offer bibliographic organization capabilities, they do not provide real-time validation of citation authenticity. Commercial hallucination detection services, though increasingly available, often impose restrictive usage limits on free tiers or require substantial subscription fees. The proposed tool fills this gap by employing a cascading validation architecture with string similarity algorithms to compute multi-dimensional match confidence scores, delivering instant feedback on reference authenticity. The system supports both single-reference verification and batch processing of BibTeX entries through a unified interface, returning validated APA citations and exportable BibTeX records within seconds.

Subjects: **Computation and Language (cs.CL)**; Computers and Society (cs.CY)

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This paper introduces CheckIfExist, an open-source tool for real-time validation of bibliographic references against multiple scholarly databases, designed to address the growing problem of citation hallucinations in AI-generated academic content.¹

